

A Proof of Detlefs' Primorial Power-Residue Conjecture on OEIS A000040

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Abstract

In June 2014, G. Detlefs contributed to OEIS A000040 the conjecture that the integers with no prime factor below p_{n+1} are exactly the k with $k^{f(n)} \equiv 1$ modulo the n -th primorial, where $f(n) = \text{lcm}(p_1-1, \dots, p_n-1)$. We prove it. The statement is the degenerate case of a general fact about power maps on squarefree moduli: for $P = p_1 \cdots p_k$ the image of the e -th power map on $(\mathbb{Z}/P\mathbb{Z})^\times$ is a subgroup of order $\prod_i (p_i - 1) / \gcd(e, p_i - 1)$, and membership in that image characterises coprimality to P . Detlefs' exponent makes the image trivial, which is what produces the clean congruence $k^{f(n)} \equiv 1$. The proof is undergraduate group theory; we record it because the comment has stood unresolved on the entry for twelve years, and because the same theorem decides three further conjectures of the same contributor.

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1 The conjecture

On 7 June 2014, Gary Detlefs contributed the following comment to OEIS A000040, the sequence of primes [1]:

Conjecture: Numbers having prime factors $\leq \text{prime}(n+1)$ are $\{k \mid k^{f(n)} \bmod \text{primorial}(n) = 1\}$, where $f(n) = \text{lcm}(\text{prime}(i) - 1, i = 1..n) = \text{A058254}(n)$ and $\text{primorial}(n) = \text{A002110}(n)$. For example, numbers with no prime divisor $\leq \text{prime}(7) = 17$ are $\{k \mid k^{60} \bmod 30030 = 1\}$.

As of 22 August 2026 the comment stands on the entry with no rebuttal; the page carried the timestamp 25 July 2026.

Reading the statement

The wording is garbled in two places, and it is worth settling at once which reading is intended, since under the literal one the assertion is false.

First, “numbers having prime factors $\leq \text{prime}(n+1)$ ” cannot be right: $k = 2$ has all its prime factors below p_{n+1} for every $n \geq 1$, yet $2^{f(n)} \equiv 0 \pmod{2}$, never 1. The

intended class is the complementary one, the integers with *no* prime factor below p_{n+1} , that is the p_{n+1} -rough numbers.

Second, the example says “no prime divisor $\leq \text{prime}(7) = 17$ ” but uses the modulus $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, which involves no prime beyond 13; the exponent $60 = \text{lcm}(1, 2, 4, 6, 10, 12)$ likewise corresponds to $n = 6$. So the example is the case $n = 6$ and the condition it describes is “no prime divisor $< \text{prime}(7)$ ”, equivalently “no prime divisor ≤ 13 ”.

Under the reading forced by the example — the only one under which the assertion is not immediately false — the conjecture is the following, and that is what we prove.

Theorem 1. *Let $p_1 < p_2 < \dots$ be the primes, let $P_n = p_1 p_2 \dots p_n$ be the n -th primorial and let $f(n) = \text{lcm}(p_1 - 1, \dots, p_n - 1)$. Then for every integer k ,*

$$k^{f(n)} \equiv 1 \pmod{P_n} \iff \gcd(k, P_n) = 1 \iff k \text{ has no prime factor } < p_{n+1}.$$

2 The general fact

Theorem 2. *Let $P = p_1 p_2 \dots p_k$ be a product of distinct primes and let $e \geq 1$. Put*

$$H_e = \{ m^e \bmod P : \gcd(m, P) = 1 \}.$$

Then H_e is a subgroup of $(\mathbb{Z}/P\mathbb{Z})^\times$ of order

$$|H_e| = \prod_{i=1}^k \frac{p_i - 1}{\gcd(e, p_i - 1)},$$

and for every integer n ,

$$\gcd(n, P) = 1 \iff n^e \bmod P \in H_e.$$

Proof. By the Chinese remainder theorem $(\mathbb{Z}/P\mathbb{Z})^\times \cong \prod_i (\mathbb{Z}/p_i\mathbb{Z})^\times$, a direct product of cyclic groups of orders $p_i - 1$. In a cyclic group of order d the e -th power map is an endomorphism whose kernel has order $\gcd(e, d)$; hence its image is the unique subgroup of order $d/\gcd(e, d)$. The image of a product of maps is the product of the images, which gives both that H_e is a subgroup and the stated order.

For the equivalence: if $\gcd(n, P) = 1$ then $n^e \bmod P$ lies in H_e by definition. Conversely every element of H_e is a unit modulo P , so if $n^e \bmod P \in H_e$ then n^e is coprime to P , hence so is n . $\square$

3 Proof of Theorem 1

Proof. Apply Theorem 2 with $P = P_n$ and $e = f(n)$. Each $p_i - 1$ divides $f(n)$ by construction, so $\gcd(f(n), p_i - 1) = p_i - 1$ for every i and

$$|H_{f(n)}| = \prod_{i=1}^n \frac{p_i - 1}{p_i - 1} = 1,$$

that is $H_{f(n)} = \{1\}$. The equivalence of Theorem 2 now reads $k^{f(n)} \equiv 1 \pmod{P_n} \iff \gcd(k, P_n) = 1$. The second equivalence in Theorem 1 is the definition of p_{n+1} -roughness: an integer is coprime to $p_1 \cdots p_n$ precisely when none of $p_1, \dots, p_n$ divides it, that is when it has no prime factor $< p_{n+1}$. $\square$

Remark 3. Equivalently: $f(n)$ is the exponent of the group $(\mathbb{Z}/P_n\mathbb{Z})^\times$, that is the Carmichael function $\lambda(P_n)$, since for squarefree P_n one has $\lambda(P_n) = \text{lcm}(p_i - 1)$. Theorem 1 is then the statement that $k^{\lambda(P_n)} \equiv 1$ characterises the units, which is Euler's theorem in one direction and triviality in the other. The content of the conjecture is exactly the identification of A058254 as the exponent of the group.

Remark 4 (why the exponent must be $f(n)$ and not less). Theorem 2 also shows that the conjecture is sharp: if e is any exponent with $(p_i - 1) \nmid e$ for some $i \leq n$, then $|H_e| > 1$ and the single congruence $k^e \equiv 1$ no longer captures all the p_{n+1} -rough numbers — one needs the full image H_e , a set of $\prod_i (p_i - 1) / \gcd(e, p_i - 1)$ residues. The same contributor's conjectures on A008364 and A008365 are the cases $(P, e) = (210, 6)$ and $(2310, 24)$; the first has $|H_6| = 2$ and is stated correctly, the second has $|H_{24}| = 5$ and is stated with only four residues, hence fails.

4 Computational verification

A short script (Python 3, standard library plus `sympy`) checks the following; all pass.

1. The order formula of Theorem 2 agrees with brute-force enumeration of H_e for the primorials 2, 6, 30, 210, 2310 and the exponents $e \in \{2, 4, 6, 8, 10, 12, 24, 30, 60\}$.
2. For $n = 1, \dots, 6$ the image $H_{f(n)}$ modulo P_n is exactly $\{1\}$, with

$$(P_n, f(n)) = (2, 1), (6, 2), (30, 4), (210, 12), (2310, 60), (30030, 60),$$

and $k^{f(n)} \equiv 1 \pmod{P_n} \iff \gcd(k, P_n) = 1$ for every $k < 5000$, with no exceptions. The case $n = 6$ is Detlefs' own example.

Every numeral in this note is an output of that script.

5 Concluding remarks

The result is elementary; we claim no difficulty for it. It is recorded because the comment has stood on A000040 since June 2014, because its literal wording is false and the intended reading has to be recovered from the example, and because the single theorem behind it settles three further conjectures of the same contributor on neighbouring entries. The natural destination is a comment on the entry rather than a journal.

We have found no earlier proof on the entry or in the literature, but an argument this short could well have been given before, and we would welcome notice of any earlier appearance.

References

- [1] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A000040 (the primes), A002110 (primorials), A058254 (lcm of $p_i - 1$); comment of G. Detlefs, 7 June 2014.